

Figure 6: MATH SFT using STM method on Gemma 2 IT 2B on different levels of token filtering levels. The best in domain performance also matches peak performance on out of domain tasks : GSM8k, ARC, MBPP as well.

Table 16: STM applied on DoRA and LoRA with MBPP on Llama 3 8B Instruct.

FINE-TUNING	TI (%)	BWT(%)
LoRA	-2.23	-34.71
LoRA + $STM_{\tau=2.5}$	3.2	-0.3
DoRA	-8.03	-1.89
DoRA + $STM_{\tau=2.5}$	3.58	1.69

instructions, we also add different answer format instructions for different tasks for the inference prompt to parse model answer correctly.

In MBPP task, we combine a problem description (‘text’), an answer format instruction and testing cases (‘test_list’) into the user prompt. The generated codes will be executed with the test case to match the execution results of ground truth codes. The following TextBox demonstrates an example of MBPP data prompts:

MBPP prompt

user prompt Please refer the given test cases and generate a python function for my problem. Make sure the written code is wrapped in code block : ```python
 < yourcode >```
 >>> Problem:{text}>>> Test Cases:{test_list}

In MATH task, we add a prefix instruction before the problem description (‘problem’) as the inference prompt. The answer will be parsed from the generated output using format of ‘\$ANSWER’. The following TextBox demonstrates an example of MATH data prompts:

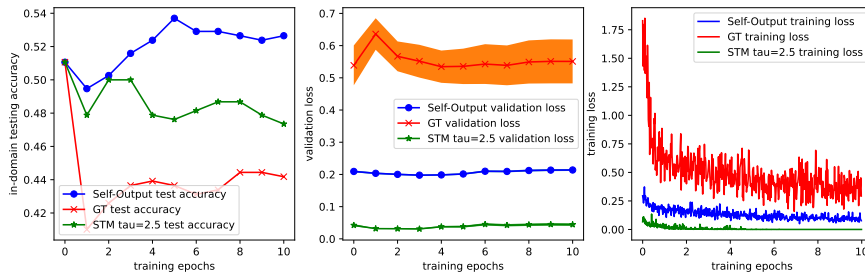


Figure 7: MBPP target task testing accuracy, validation and training loss of ground truth, Self-Output, and STM data for Gemma 2 IT 2B.

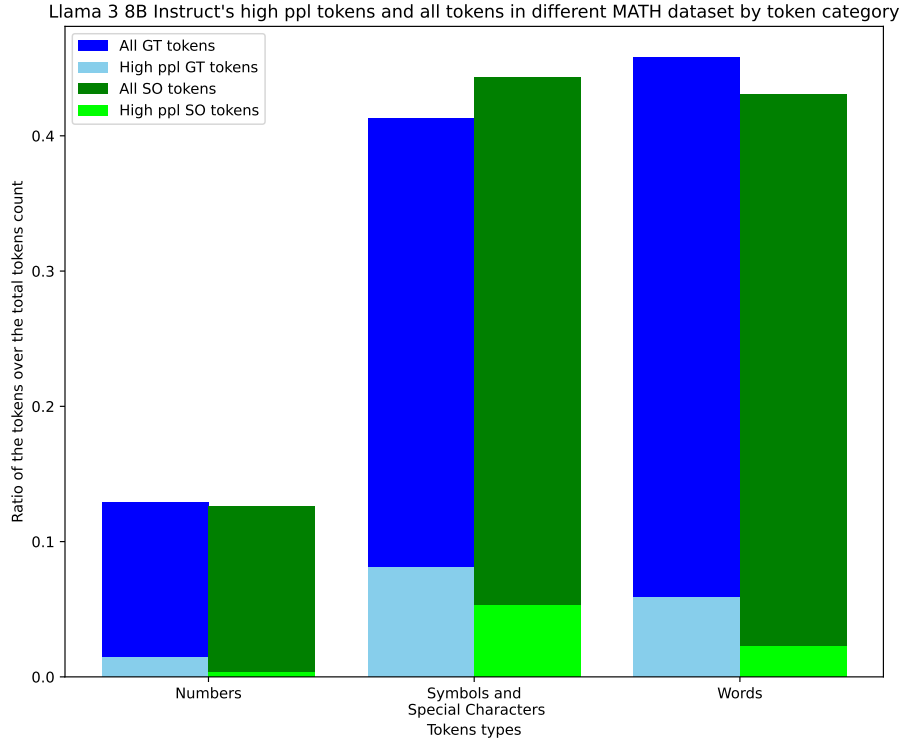


Figure 8: Ratio changes of high perplexity tokens before and after STM filtering on ground-truth (GT)dataset and Self-Output (SO) dataset of MATH task with Llama 3 8B Instruct

Table 17: Comparison of efficiency between STM, Self-Output and Rephrase in terms of gpu hours on MBPP target task.

MODEL NAME	NUMBER OF SAMPLE	CORRECTNESS	GPU HOURS
BASELINE FINE-TUNING	374	100%	0
STM LLAMA 3 8B INSTRUCT	374	100%	4.5 MINUTES
SELF-OUTPUT LLAMA 3 8B INSTRUCT	213	57%	16.8 HOURS
REPHRASE LLAMA 3 8B INSTRUCT	327	87.4%	36.8 MINUTES

MATH prompt

user prompt Solve the following math problem step by step. The last line of your response should be of the form Answer: \$ANSWER (without quotes) where \$ANSWER is the answer to the problem.
problem
Remember to put your answer on its own line after "Answer:", and you do not need to use a boxed command.

For Non-target task: In ARC-Challenge task, we add a Question-Answering task instruction and an answer format instruction before the ‘question’ and answer choices (‘choices’) as the user role prompt, and directly use the ‘answerKey’ value to match with the parsed results from the output. The TextBox shows the user prompt is like:

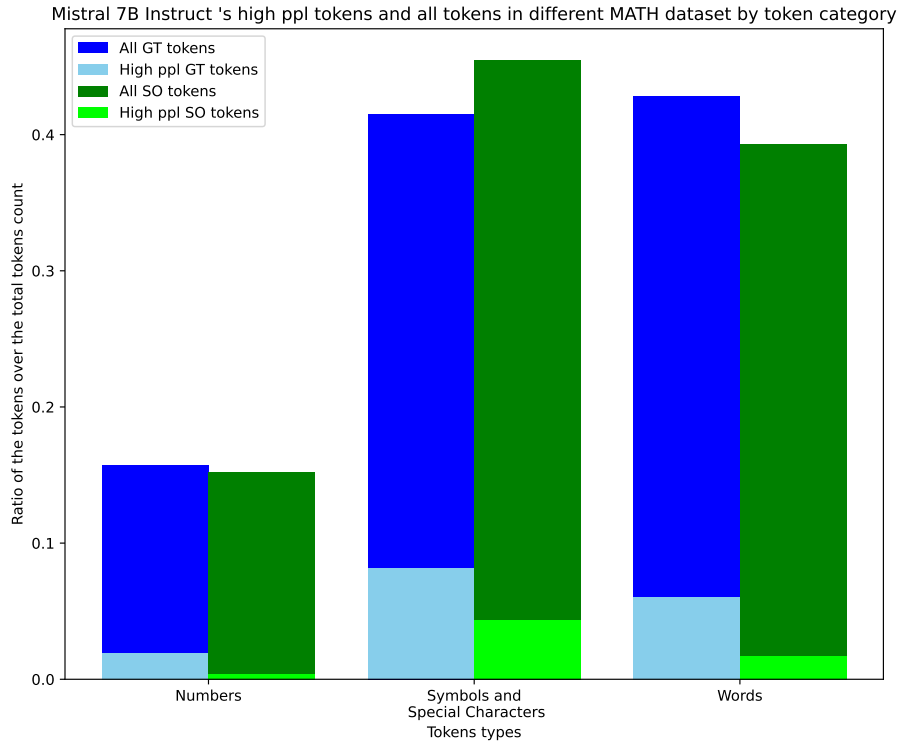


Figure 9: Ratio changes of high perplexity tokens before and after STM filtering on ground-truth (GT) dataset and Self-Output (SO) dataset of MATH task with Mistral 7B Instruct

ARC Challenge prompt

user prompt Answer the following multiple choice question. The last line of your response should be of the following format: 'Answer: \$LETTER' (without quotes) where LETTER is one of ABCD.

Question: {question}

Choices: {choices}

In GSM8k task, we add a Question-Answering task instruction and an answer format instruction before the 'question' as the user role prompt. To evaluate the output, we directly parse the format of '#### <number only answer>' from the generated results, and match it with the 'answer' value from the dataset. The TextBox shows an example of the user prompt:

GSM8k prompt

user prompt You are given a grade school math question. Please answer the question in the following format:

Q: <question>

A: <Think step by step here> #### <number only answer>

Format requirements : you must first output your reasoning before finalized with the "#### " format followed by the final numeric answer.